

Next-generation image captioning: A survey of methodologies and emerging challenges from transformers to Multimodal Large Language Models

Huda Diab Abdulgalil*, Otman A. Basir

Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, Ontario, Canada



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ABSTRACT

The widespread availability of visual data on the Internet has fueled a significant interest in image-to-text captioning systems. Automated image captioning remains a challenging multimodal analytics task, integrating advances in both Computer Vision (CV) and Natural Language Processing (NLP) to understand image content and generate semantically meaningful textual descriptions. Modern deep learning-based approaches have supplanted traditional approaches in image captioning, leading to more efficient and sophisticated models. The development of attention mechanisms and transformer-based architectures has further enhanced the modeling of both language and visual data. Despite these gains, challenges such as long-tailed object recognition, bias in training data, and shortcomings in evaluation metrics constrain the capabilities of current models. Furthermore, an important breakthrough has been made with the recent emergence of Multimodal Large Language Models (MLLMs). By incorporating textual and visual data, MLLMs provide improved captioning flexibility, generative capabilities, and reasoning. However, these models introduce new challenges, including faithfulness, grounding, and computational cost. Although relatively few studies have comprehensively surveyed these developments, this paper provides a thorough analysis of Transformer-based captioning approaches, investigates the shift to MLLMs, and discusses associated challenges and opportunities. We also present a performance comparison of the latest models on the MS-COCO benchmark and conclude with perspectives on potential future research directions.

1. Introduction

The enormous amount of visual information contained in images makes visual understanding techniques essential for automated tasks such as image captioning that would otherwise be challenging for humans to perform. Generating precise image descriptions is a complex task that involves integrating advancements in Computer Vision (CV) and Natural Language Processing (NLP) domains to understand the image content and translate it into semantically meaningful textual descriptions. Automated image captioning, referred to by the term “image captioning” in the literature (Salgotra et al., 2024; Xu et al., 2023; Sharma, 2023; Sharma and Padha, 2023a; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023; Tsaniya et al., 2022; Xu et al., 2022; Bucciarelli et al., 2024) involves describing visual content for a given image with a plain, semantically accurate, and natural language sentence. These frameworks require both a comprehensive visual understanding system to recognize objects and their interrelationships, in addition to a capable language model to generate meaningful and syntactically correct sentences that match visual content (Tsaniya et al., 2022; Xu et al.,

2022; Ambilduke et al., 2023; Sharma and Padha, 2023b; Ondeng et al., 2023). Image captioning holds promise for assisting individuals with visual impairments via image-to-speech systems and benefits downstream tasks such as sentiment analysis and information aggregation in applications like autonomous driving. Despite its potential, image captioning remains a subject of extensive research aimed at improving the associated algorithms.

Image captioning has evolved over time, beginning with traditional methods such as template filling, description retrieval, and manually created natural language-generating methods (Salgotra et al., 2024; Xu et al., 2023; Sharma, 2023; Sharma and Padha, 2023a; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023; Tsaniya et al., 2022). While traditional systems, such as template-based and retrieval-based models can provide pertinent captions for images, they often lack diversity and creativity, since they rely on established templates, making it difficult to generate new or contextually appropriate descriptions for unseen query images. Motivated by the success of deep learning methodologies in machine translation, deep learning-based methods have advanced the

* Corresponding author.

E-mail address: hdiababd@uwaterloo.ca (H.D. Abdulgalil).

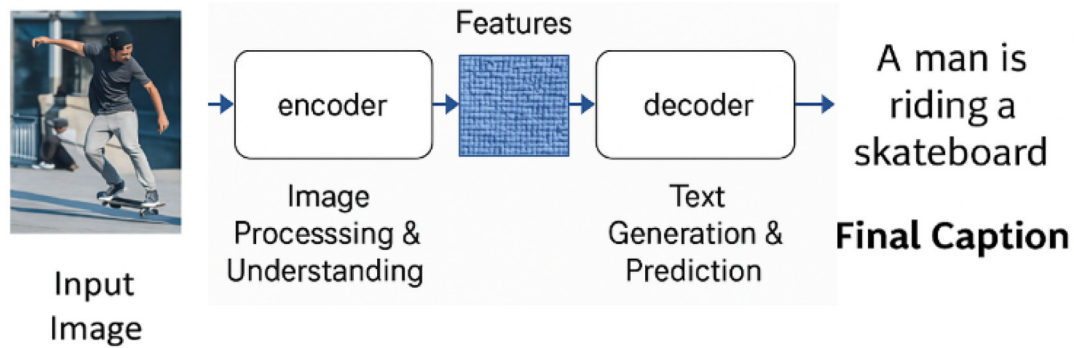


Fig. 1. The general structure framework for image captioning.

effectiveness of image captioning using the encoder–decoder architecture (Sharma and Padha, 2023b; Ondeng et al., 2023). The basic structure of the deep learning framework is shown in Fig. 1. Early deep learning-based frameworks, also known as non-attention methods, relied on CNN-based global features to retrieve high-level representations from images. In these models, a Convolution Neural Network (CNN) is utilized as an encoder to extract a set of visual characteristics, which are subsequently transmitted into a language model decoder, typically a Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) network. The decoder generates the following word based on the partially created caption and the visual data supplied by the encoder. These methods are computationally efficient and simple; however, they suffer from a loss of granularity and fine-grained details, making it difficult to generate specific, contextually rich, and detailed descriptions of image content. Additionally, global features extracted by CNNs fail to capture spatial relationships and object-level interactions, both of which are crucial for creating accurate and descriptive captions (Ambilduke et al., 2023; Sharma and Padha, 2023b; Ondeng et al., 2023). Moreover, since RNNs are recurring in nature, CNN-RNN or CNN-LSTM frameworks result in generating ambiguous and generic sentences and may fail to adequately represent the nuances of image (Tsaniya et al., 2022; Xu et al., 2022).

Advanced deep learning-based approaches for image captioning have evolved to include attention mechanisms (Xu et al., 2015) and Transformer (Vaswani et al., 2017), motivated by their exceptional performance in machine translation and NLP tasks. The development time frame for image captioning techniques is presented in Fig. 2. These techniques enable more accurate and contextually relevant captions by allowing the model to concentrate dynamically on the most relevant regions of the image during the caption production process. Since its original introduction in 2017 (Vaswani et al., 2017), the transformer model has served as a solid basis for numerous modern models (Luo et al., 2021; Xian et al., 2022b; Nguyen et al., 2022; Dosovitskiy et al., 2020; Guo et al., 2020; Ayad, 2024). Transformer has been progressively included in almost all multimodal tasks, including image captioning (Luo et al., 2021; Xian et al., 2022b; Nguyen et al., 2022; Dosovitskiy et al., 2020; Guo et al., 2020; Pan et al., 2020; Cornia et al., 2020; Li et al., 2019; Herdade et al., 2019; He et al., 2020; Zhang et al., 2021b; Radford et al., 2021; Yu et al., 2019; Li et al., 2020; Hu et al., 2021; Lu et al., 2023; Zheng et al., 2023; Li et al., 2024; Abdal Hafeth and Kollias, 2024; Mundu et al., 2024; Basak et al., 2024; Bui et al., 2024; Chen and Li, 2025; Xu et al., 2024; Huang and Li, 2022; Fan et al., 2021; Liu et al., 2021b; Jiang et al., 2022). Unlike RNNs and LSTMs, which process input sequences sequentially and can struggle with long-range dependencies (Stefanini et al., 2022), Transformers process input sequences in parallel (Stefanini et al., 2022). Furthermore, Transformers use positional encoding to model the sequence of input elements, which allows them to better capture spatial relationships that are essential for accurate caption creation compared to other architectures (Stefanini et al., 2022). Their inherent scalability and

architectural adaptability make Transformers particularly well-suited for handling complicated tasks like image captioning, where it is crucial to capture minor and fine-grained interactions between visual inputs and language output. Transformers have become the industry standard in recent years, not only for image captioning but also for other applications like medical image analysis, where accuracy and resilience are critical (Sakirin and Said, 2025; Lou et al., 2022; Zhu et al., 2022; Hu et al., 2022c; Lou et al., 2023; Guo et al., 2023; Wang et al., 2023; Parvin et al., 2023; Elbedwehy et al., 2022; Dandwate et al., 2023b; Lin et al., 2014), which can discover the correlations between visual characteristics and enable parallel training of the sequences.

Despite these advances, Transformer-based frameworks still face limitations in producing fully accurate, contextually rich captions (Xu et al., 2024; Huang and Li, 2022; Fan et al., 2021; Jiang et al., 2022; Wu et al., 2022b; Lou et al., 2022; Zhu et al., 2022; Hu et al., 2022c; Lou et al., 2023; Guo et al., 2023; Wang et al., 2023; Parvin et al., 2023; Elbedwehy et al., 2022; Dandwate et al., 2023b; Tan et al., 2022). Balancing the memory-intensive demands of global self-attention against the real-time constraints of inference remains a constant trade-off. Rare and long-tail object occurrences still require expensive retraining or data augmentation. A major limitation of current transformer-based models lies in their primary reliance on single features like regional or grid characteristics as their standard visual information representation, resulting in inadequate global context, misdetection objects, and high computational overhead. These methods have difficulty managing interactions between several visual features because they frequently depict internal connections in a single feature while ignoring the advantageous effects of using multiple features to obtain more superior feature representation during the encoding stage. This lack of integration impedes the visual information flow during the decoding phase, forcing the language decoder module to heavily rely on partially formed captions to guide the generation process. This results in the creation of erroneous words in the subsequent time step, making the generated descriptions unrelated to the visual content and reducing the quality of captions. Recent efforts have begun fusing grid and regional characteristics to address these challenges (Yu et al., 2022; Wang et al., 2022c; Li et al., 2023); however, optimal fusion strategies are still unreachable. Alternative backbone paradigms, such as Perceiver IO's cross-modal bottleneck layers, offer leaner architectures that substantially reduce computation and memory requirements but slightly lower peak accuracy. Recent advancements in Multimodal Large Language Models (MLLMs) have also integrated Vision Transformers with large-scale pre-trained models to better handle the complexity of generating rich, detailed captions (Bucciarelli et al., 2024).

A significant breakthrough in vision–language understanding has been achieved with the emergence of Multimodal Large Language Models (MLLMs), such as LLaVA (Liu et al., 2024) and IDEFICS (Laurençon et al., 2023). These models are capable of processing both textual and visual inputs simultaneously, generating captions with varying levels

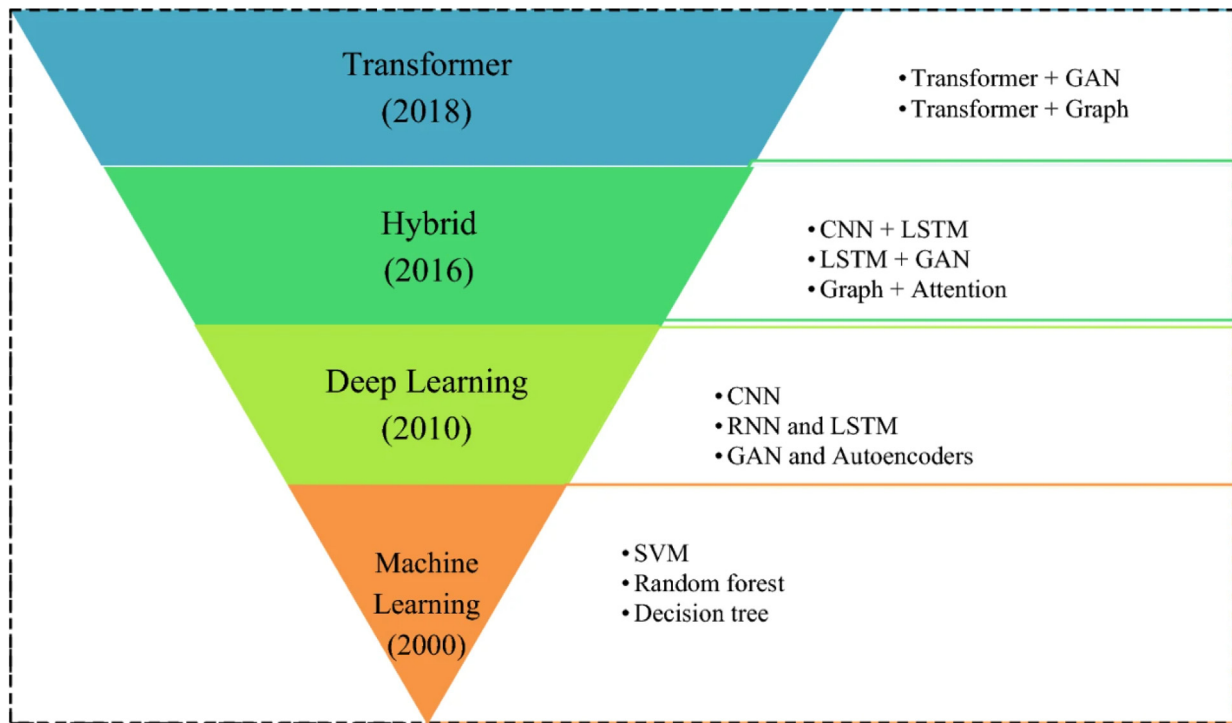


Fig. 2. A progressive development of the image captioning system (Salgotra et al., 2024).

of abstraction, specificity, and length. MLLMs represent a natural progression from traditional image captioning models, offering enhanced reasoning and generative capabilities. Positioned at the cutting edge of current research, these models significantly influence the evolution of the field. However, their introduction has also highlighted the need for more precise, contextually aware evaluation methods that can assess the intricacy, depth, and accuracy of captions produced by MLLMs. This shift has also led to new challenges in evaluating the quality of generated captions, as current metrics, such as BLEU, CIDEr, and METEOR, may fail to capture the multi-dimensional and complex nature of captions produced by MLLMs. Despite the remarkable progress of these models, they introduce unique challenges, including hallucinations, content faithfulness, and increased computational costs compared to traditional approaches.

This paper offers a comprehensive review of image captioning models, emphasizing recent advancements, particularly those incorporating Transformers and the emerging MLLMs. While earlier reviews predominantly centered on deep learning techniques such as LSTMs (Liu et al., 2023; Xu et al., 2023; Sharma, 2023; Sharma and Padha, 2023a; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022) and traditional attention mechanisms (Khaing, 2019; Osman et al., 2023; Li et al., 2024; Abdal Hafeth and Kollias, 2024) often neglect transformers and other recent advancements. Some studies, such as (Verma and Kene, 2023; Wang et al., 2019b; Alsayed et al., 2023), examine transformers as vision models and as language models but do not concentrate on their role in image captioning. This paper fills this gap by reviewing Transformer-based systems and the rising importance of MLLMs. While prior reviews, including (Tsaniya et al., 2022; Ambilduke et al., 2023; Ondeng et al., 2023), have categorized transformer-based methodologies into two categories: vanilla models based on the initial transformer and Vision-Language Pre-Training (VLP) models based on recent advances in pre-training. In contrast, beyond the usual categories, this paper introduces two new categories: transformer approaches based on single features and transformer approaches based on fusion/hybrid features, which have not been fully explored in previous review papers, providing a more sophisticated view of image captioning systems. This paper reviews a broad range

of image captioning approaches published between 2019 and 2024. A comparison of the various models is also highlighted based on the MS-COCO dataset (Lin et al., 2014) using “Karpathy” split (Karpathy and Fei-Fei, 2015). This study contributes to the understanding of transformers’ potential in image captioning, while also considering MLLMs as the obvious next development in image captioning methodologies. Other contributions can be summarized as follows:

- Conduct a comprehensive analysis of current automated image captioning models, with a focus on transformer-based approaches.
- Compare the performance of the latest transformer-based models using the benchmark MS-COCO image-caption dataset.
- Discuss the limitations of existing transformer-based approaches, while exploring potential advancements, including insights into MLLMs and their ability to enhance and complement current transformer-based methods.

The rest of this paper is organized as follows: Section 2 covers advances in image captioning, with emphasis on transformer-based models. Section 3 delves into visual feature representation methodologies in the context of image captioning. In Section 4, existing approaches are highlighted and categorized based on the innovation type including, Attention Mechanisms, Fusion Strategies, Pre-training Approaches, and MLLMs. Section 5 discusses the limitations of these methods, followed by Section 6, which provides a comparative analysis of the performance of various models, reviews recent advancements, and explores the emerging role of Multimodal Large Language Models (MLLMs). Section 7 outlines the remaining challenges and potential future directions in the field of image captioning. Finally, Section 8 concludes the study.

2. Image captioning methodologies

In the early stages of image captioning research, retrieval-based and template-based approaches were primarily used to generate a coherent and comprehensible phrase for the given image. However, most cutting-edge methods for image captioning now rely heavily on deep learning

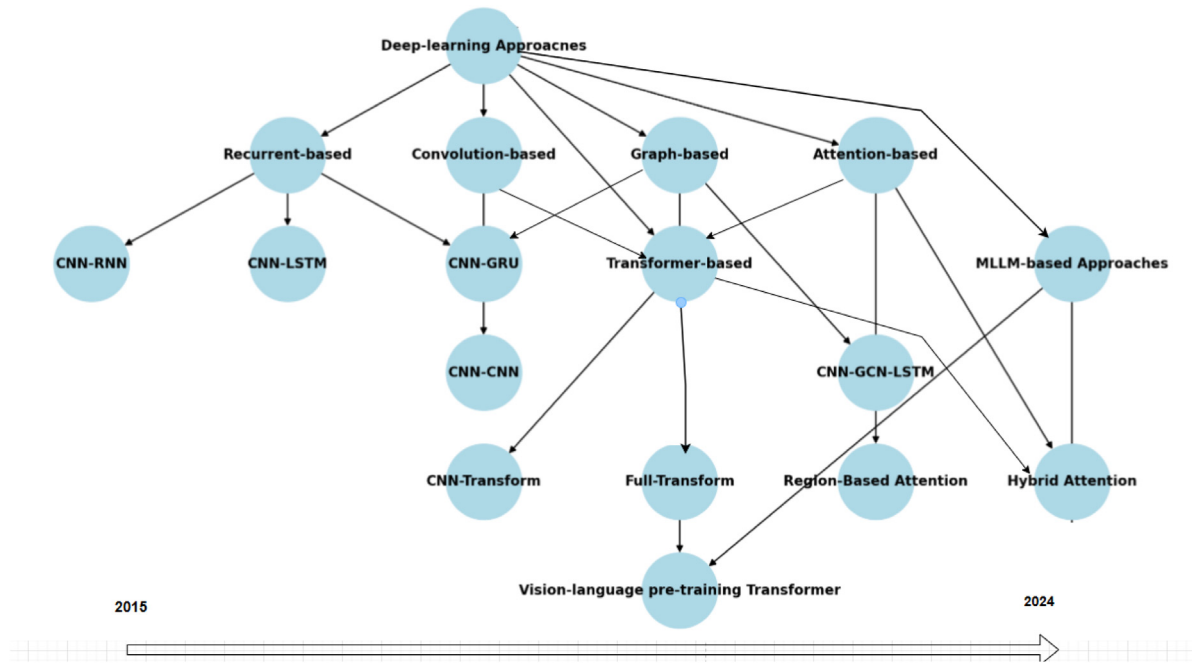


Fig. 3. Deep-learning paradigm shift in automated image captioning: From Recurrent Models to MLLMs.

techniques, which can produce captions that are more semantically precise than traditional methods (Xu et al., 2022; Ambilduke et al., 2023; Ondeng et al., 2023; Salgotra et al., 2024). An image caption framework based on deep learning involves three basic processes, as classified in Stefanini et al. (2022), including visual encoding models, training strategies, and language decoding models. The common training strategies used in deep learning-based image captioning include cross-entropy loss, vision and language pre-trained objectives, and reinforcement learning. The majority of models in the literature adhere to this standard design pattern (Xu et al., 2023; Sharma, 2023; Sharma and Padha, 2023a; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023; Tsaniya et al., 2022; Xu et al., 2022). Although we briefly mention traditional retrieval and template methods, this paper focuses on deep-learning approaches, especially transformer-based architectures, and the recent emergence of MLLM-based pipelines. The evolution and the switch in automated image captioning algorithms based on deep-learning techniques from recurrent and convolutional models through attention and transformer-based systems to MLLM-based is illustrated in Fig. 3. The encoder-decoder framework, originally derived from machine translation models, forms the foundation of most image captioning models. This framework frequently incorporates attention techniques, which have significantly enhanced model performance by enabling dynamic focus on relevant image regions or sequence elements of the generated sequence. While the encoder-decoder structure establishes the information flow, the attention mechanism dynamically guides the model to prioritize the most crucial aspects at each stage during caption generation (Sharma and Padha, 2023a; Ming et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023). The encoder should be capable of learning and effectively representing the visual content. CNNs have historically served as the foundation of image feature extractors. However, with the advent of Vision Transformers (ViTs) in 2022, the role of image feature extraction has shifted, as ViTs now offer a fully transformer-based approach to visual encoding (Alsayed et al., 2023). ViTs offer advantages over CNNs, particularly in capturing long-range dependencies within the image. The decoder estimates the probability of a word sequence using language modeling techniques, with RNNs traditionally being the most common choice. Xu et al. (2023), Sharma (2023), Ming et al. (2022), Stefanini et al. (2022), Ghandi et al.

(2022), Kavitha and Karpagam (2022), Osman et al. (2023), Tsaniya et al. (2022). However, attention-based and transformer decoders have gained popularity due to their improved performance in sequence prediction tasks. Due to the shortcomings of RNNs and LSTMs as language models, researchers have turned to substitutes like Transformers and MLLMs, which fundamentally altered the process of language creation. Transformers use fixed or learned positional encodings to maintain the order of input elements while operating in parallel, in contrast to RNNs and LSTMs that process inputs sequentially. To create captions that are both logical and contextually accurate, the model must be able to store sequence information, which is made possible by these positioning cues. For complex visual-linguistic tasks like image captioning, Transformer-based models are especially effective because they use self-attention mechanisms to capture global context across the whole input (Tsaniya et al., 2022)(Alsayed et al., 2023). Large-scale pre-trained models and ViTs have recently been incorporated into MLLMs to tackle the complexities of generating rich, detailed captions. MLLMs leverage the advantages of both transformer-based visual encoding and the rich language generation capabilities of transformers to improve image captioning accuracy (Bucciarelli et al., 2024). Recent improvements in image captioning algorithms, particularly transformer-based models, have illustrated their effectiveness in natural language generation tasks (Tsaniya et al., 2022; Xu et al., 2022; Ambilduke et al., 2023). In contrast to the other categories, the transformer-based models are the most prominent variation of the attention, with a significant interest in their application (Tsaniya et al., 2022; Xu et al., 2022; Ambilduke et al., 2023; Sharma and Padha, 2023b; Ondeng et al., 2023). Deep learning-based image captioning approaches have been extensively studied in previous reviews. Numerous reviews on deep learning methods for image captioning have been conducted (Salgotra et al., 2024; Xu et al., 2023; Sharma, 2023; Sharma and Padha, 2023a; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022). Khaing and Phyu (Khaing, 2019) presented a comparison investigation for attention-based approaches. Despite the advancements brought about by transformers, current models still face several limitations (Tsaniya et al., 2022; Xu et al., 2022; Ambilduke et al., 2023). There is limited literature covering the advances in image captioning, especially with a focus on the model domain, namely, the transformer models.

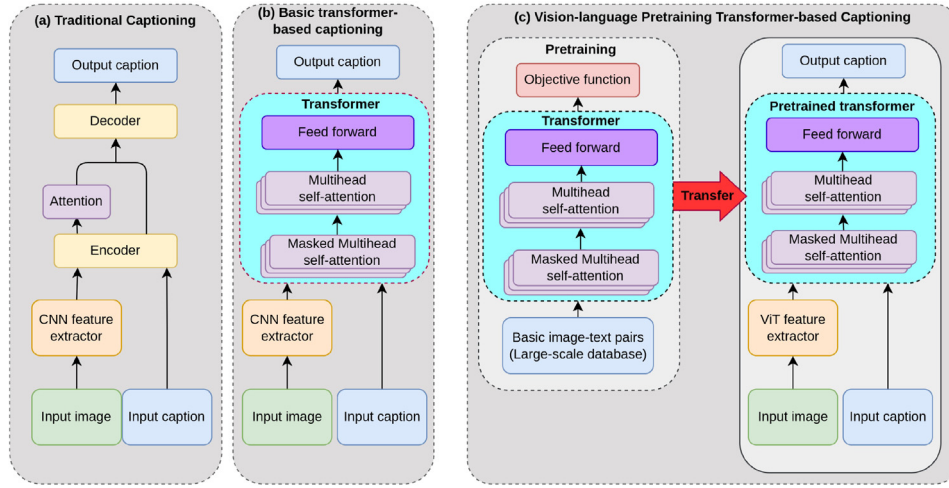


Fig. 4. Development of captioning structures and systems (Ondeng et al., 2023).

Transformer-based methodologies are often divided into two categories: vanilla models based on the original transformer and Vision-Language Pre-Training (VLP) models based on recent advances in pre-training (Ambilduke et al., 2023; Sharma and Padha, 2023b; Ondeng et al., 2023). Fig. 4 depicts the development of captioning structures and systems, from (a) the earliest encoder–decoder methods to (b) the vanilla transformer-based captioning methods towards (c) the transformer-based captioning methods that include VLP. Along with these primary categories, this paper introduces two new categories, transformer approaches based on single features and transformer approaches based on fusion/hybrid features, providing a more detailed classification of the recent advancements in image captioning.

2.1. Standard image captioning evaluation metrics

In image captioning research, evaluation metrics such as BLEU, SPICE, CIDEr, METEOR, and ROUGE serve as critical benchmarks for assessing model performance. These metrics are essential for measuring the accuracy and quality of linguistic descriptions, allowing for effective comparison and evaluation of various models. A brief overview of each of these standard evaluation metrics is presented in the following subsection and summarized in Table 1.

2.1.1. BLEU metric

BLEU is an abbreviation of Bilingual Evaluation Understudy (Papineni et al., 2002), which widely used to evaluate machine translation systems (Xu et al., 2023; Sharma, 2023; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023). The BLEU score measures n -gram precision between a candidate caption and references while penalizing overly short outputs (Sharma and Padha, 2023a). BLEU is calculated as shown in Table 1. Given an image denoted as I and its associated annotations denoted as S , where s_i stands for the sentence's i th word. The produced caption by the model is denoted as $c_i \in C$, in which C is the collection of all c_i . The BLEU score is often computed for various values of N (typically 1 to 4) to capture different n -gram co-occurrences. For BLEU-4, the n -grams of all four levels are determined, and each is assigned a weight of 0.25. The term w_n represents the weight of each n -gram accuracy. BP is a penalty factor, and P_n is the matching precision. The BLEU final score is equal to the logarithmic weighted average times BP , which calculated as follows: if the candidate length l_c is greater than the reference length l_s , then $BP = 1$; otherwise, $BP = \exp\left(1 - \frac{l_s}{l_c}\right)$, where the lengths of the produced and reference sentences are denoted by l_c , l_s , respectively. Further details can be found in Papineni et al. (2002), Sharma and Padha (2023a).

2.1.2. METEOR metric

METEOR is an abbreviation of Metric for Evaluation of Translation with Explicit Ordering (Banerjee and Lavie, 2005), commonly used for machine translation systems. METEOR evaluates captions by computing a weighted F-score over exact, stem, and synonym matches, penalizing disordered phrasing to better reflect human judgment of recall and precision (Stefanini et al., 2022). METEOR metrics use matching unigram along with a penalty function for erroneous word arrangement to determine the accuracy, recall rate, and weighted F-score for every instance of a corresponding word, root, and equivalent (Banerjee and Lavie, 2005). The METEOR score is calculated as shown in Table 1, where F_{mean} is the harmonic mean of recall and precision, and Pen is the penalized coefficient. Precision is determined by $P = \frac{|m|}{\sum_k h_k(c_i)}$, and recall is determined by $R = \frac{|m|}{\sum_k h_k(s_{ij})}$, where c denotes how many pieces are matched and m denotes the overall number of matches. While it captures linguistic variation more robustly than BLEU, it still relies on reference overlap and can be slow to compute. More details can be found in Banerjee and Lavie (2005), Stefanini et al. (2022), Sharma and Padha (2023a).

2.1.3. ROUGE metric

ROUGE is an abbreviation of Recall Oriented Understudy for Gisting Evaluation (Lin, 2004), which was developed for automated text summarization. The fundamental concept of ROUGE comprises the counting of word pairings, word cycles, and n -grams to assess the efficacy of abstracts. In this metric, the sequence of words as well as word pairings are compared to reference descriptions. ROUGE appears in several forms, including ROUGE-N, ROUGE-S, and ROUGE-L (Ming et al., 2022; Sharma, 2023). The ROUGE-L score is calculated as shown in Table 1. In which the recall is calculated as $R_{les} = \frac{LCS(C,S)}{m}$, and the precision is calculated as $P_{les} = \frac{LCS(C,S)}{n}$ where m and n represent the lengths of the generated and reference sentences, respectively. While ROUGE is effective for evaluating the sufficiency and accuracy of a caption, it does not measure fluency or semantic coherence (Sharma, 2023; Ming et al., 2022).

2.1.4. SPICE metric

SPICE stands for Semantic Propositional Image Captioning Evaluation (Anderson et al., 2016), utilizes a graph-based semantic schema to evaluate captions. Using the corresponding descriptions of the images, this graph can be employed to extract data from multiple objects, their attributes, and the connections between them. In contrast to conventional metrics, SPICE gives semantic content precedence over fluency (Anderson et al., 2016) since it is based on matching tuples derived from scene graphs, which are produced by the reference and

Table 1
Standard evaluation metrics for image captioning.

Metric	Description	Formula
BLEU@N (Papineni et al., 2002)	Measures n-gram precision between predicted and reference captions.	$BLEU@N = BP \times \exp\left(\sum_{n=1}^N w_n \log P_n\right)$
METEOR (Banerjee and Lavie, 2005)	Incorporates precision, recall, stemming, and synonym matching for richer alignment.	$METEOR = (1 - Pen) F_{mean}$
ROUGE-L (Lin, 2004)	Computes the longest common-subsequence recall-oriented matching between captions.	$ROUGE-L = \frac{(1 + \beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}}$
SPICE (Anderson et al., 2016)	Uses scene-graph tuples to focus on semantic content rather than surface overlap.	$SPICE = \frac{2 P(C, S) R(C, S)}{P(C, S) + R(C, S)}$
CIDEr (Vedantam et al., 2015)	Weights n-gram consensus by TF-IDF to capture human-like agreement.	$CIDEr = \frac{1}{N} \sum_{n=1}^N CIDEr_n(C_i, S_i)$

the candidate. SPICE score is calculated as shown in Table 1 where $R(C, S)$ represents recall and $P(C, S)$ represents precision. Similar to METEOR, SPICE searches WordNet for synonyms and considers them to be good matches. This makes it more aligned with human perception of meaning, though it depends on the quality of scene-graph parsers (Ming et al., 2022; Osman et al., 2023).

2.1.5. CIDEr metric

CIDEr stands for Consensus-based Image Description Evaluation (Vedantam et al., 2015). This metric has been specifically designed for the evaluation of image captions (Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023). This metric measures the percentage of candidate statement n-grams in the reference statement, determined using the TF-IDF (Term frequency-inverse document frequency) method, which also calculates the average cosine distance between reference and model captions to assess their similarity. CIDEr score is calculated as shown in Table 1, where C_i represents the produced caption and S_i is the reference caption. Compared to BLEU, METEOR, and ROUGE, CIDEr is less affected by shorter phrases. SPICE and CIDEr have a greater correlation with human judgments than the other metrics, but can be biased by reference set size and domain vocabulary. Sharma and Padha (2023a), Ghandi et al. (2022), Kavitha and Karpagam (2022). More details can be found in Vedantam et al. (2015).

2.2. Recent developments in image captioning evaluation metrics

The n-gram overlap with reference captions was the subject of early metrics like BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), and METEOR (Banerjee and Lavie, 2005), which were initially created for machine translation. Although these metrics were widely employed, their dependence on language overlap and reference captions limited their effectiveness. Consensus-based scoring and scene graph analysis were included in later metrics like CIDEr (Vedantam et al., 2015) and SPICE (Anderson et al., 2016) to increase correlation with human judgment. Even with these improvements, these measurements still mostly rely on textual comparisons, struggle to match synonyms, and assume that reference captions fully express the image content. In an attempt to gain control over these limitations, the focus of advancements has turned to learnable metrics that use pre-trained models in language and vision, such as HiFi-S (Yao et al., 2024), and CLIP (Mokady et al., 2021). These metrics can capture hierarchical structures and localized meanings; however, they also come with their inherent restrictions (Sarto et al., 2025). For instance, due to the limitations in training data quality, large-scale models like CLIP, which were trained on web-sourced data, may not be fully effective for image caption evaluation. Multimodal models and other components are being incorporated into recent evaluation metrics to boost performance, specifically by enhancing their capacity for recognizing fine-grained features. Although early attempts used BERT-based embeddings to compare generated sentences with reference captions, more current methods make use

of Multimodal Large Language Models' sophisticated reasoning and broad pre-training capabilities for more reliable assessment scores. For example, HiFi-S (Yao et al., 2024) demonstrates the growing acceptance of MLLMs in caption evaluation in recent years. As MLLMs continue to evolve, their capacity to produce a wider variety of captions poses additional difficulties for traditional evaluation criteria, which might not be suitable to accurately assess the overall quality of generated captions (Sarto et al., 2025).

2.3. Image captioning datasets

High-quality datasets are critical for the development and evaluation of image captioning models. Early image captioning architectures (Vinyals et al., 2015; Xu et al., 2015) were commonly trained and tested on the Flickr8K (Young et al., 2014) and Flickr30K (Plummer et al., 2015) datasets, collections of real-world images sourced from Flickr, each annotated with five descriptive captions. The Microsoft Common Objects in Context (MS-COCO) dataset (Lin et al., 2014), which comprises diverse everyday scenes involving people, animals, and objects in context, has become the dominant benchmark in the field (Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023). The design and functionality of image caption approaches are greatly influenced by the attributes of the available captions, in addition to their quantity (Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023). Most recent studies adopt the MS-COCO dataset with the training, validation, and test splits proposed by Karpathy and Fei-Fei (2015). This paper primarily focuses on Transformer-based models and recent approaches evaluated on the MS-COCO dataset. A summary of commonly used datasets is presented in Table 2.

A main drawback of these widely used benchmarks is their notable domain bias. While commonplace scenes predominate in these collections, specialized domains such as medical, industrial, or fine-art images are undersampled, which can lead models to produce generic, stereotyped captions. MS-COCO captions tend to be shorter and more homogeneous, yielding models that excel at general scene description but may struggle with fine-grained language. In contrast, Flickr30K captions are longer and more detailed, encouraging models to learn complex sentence structures and richer semantics. Empirically, MS-COCO trained systems often post slightly higher aggregate scores; likely a by-product of the dataset's consistency, research, whereas Flickr30K models may generalize better to varied or syntactically complex captions.

3. Feature representation methodologies in image captioning

In the image captioning framework, determining the right visual feature representation approach is essential for producing precise descriptions. In the literature, the representation of visual elements has evolved through three major stages, including grid-based, region-based, and fusion/hybrid-based. A brief description of these stages, along with their advantages and disadvantages, is provided in this section.

Table 2
Summary of common image captioning datasets.

Dataset	Number of images	Number of captions per image	Number of words per caption	Vocabulary size	Advantages
MSCOCO (Lin et al., 2014)	132 K	5	10.5	27 K	Large, diverse dataset covering a wide range of real-world scenes.
Flickr30K (Plummer et al., 2015)	31 K	5	12.4	18 K	Various daily activities, excellent for testing generalization.
Flickr8K (Young et al., 2014)	8 K	5	10.9	8 K	Small and manageable, ideal for rapid prototyping and debugging.

3.1. Regional features

In the early stages of image captioning, region-based features were commonly used to capture visual information. To discover visual information from images, researchers first employed a pre-trained CNN detector on ImageNet (Russakovsky et al., 2015) for encoding. Motivated by human vision pattern, Anderson et al. (2018) first extracted a set of regional features that can offer object-level detail using Faster R-CNN (Ren et al., 2015). Subsequently, regional characteristics have emerged in the first stage as the accepted method of visual representation for many image caption approaches. This paradigm has become a standard pattern for extracting image features, which was extensively adopted in later research (Jiang et al., 2020; Pan et al., 2020; Cornia et al., 2020; Li et al., 2019) because of its evident benefits in line with human visual habits. However, regional characteristics come with their own set of drawbacks, such as the absence of comprehensive semantics and context (Ming et al., 2022; Osman et al., 2023). Furthermore, since collecting region-level features takes time, the majority of current models are directly trained and examined on cached data rather than images, making it challenging to train an image captioning model end-to-end (Tsaniya et al., 2022; Xu et al., 2022; Ambilduke et al., 2023; Sharma and Padha, 2023b; Ondeng et al., 2023). To address this, several studies (Wang et al., 2022a; Ma et al., 2023; Jiang et al., 2020) used attention processes in conjunction with finer-grained grid-based characteristics to represent raw images and better generate captions by utilizing granularity. Additionally, while Faster R-CNN pre-trained on large datasets offers solid performance, its reliance on pre-training and absence of fine-grained information make it a costly and time-consuming process (Nguyen et al., 2022). While region characteristics provide accurate object localization, they come at a significant computational cost and eliminate comprehensive scene understanding.

3.2. Grid features

Grid features can better capture contextual information and comprehensive semantics since they encompass all of an image's material in a more fragmented manner. Grid characteristics capture the entire image via a uniform patch or feature-map grid, using backbones like VGG-16 or ResNet-101 (Simonyan and Zisserman, 2014; He et al., 2016). These features lack precise object delineations but are generally more information-rich than region proposals. A high-layer characteristics representation within the backbone network, such as CNNs or vision-based Transformers (ViTs), is usually applied to obtain grid features from the entire image (Nguyen et al., 2022). Many studies (Wu et al., 2022; Rennie et al., 2017; Zhang et al., 2021b) employ CNNs that have been pre-trained on the dataset with VGG-16 (Simonyan and Zisserman, 2014) and ResNet-101 (He et al., 2016) to serve as encoders to convert images into fixed-length grid-level features directly from input images. Grid features generally offer better performance and lower latency compared to region features (Tsaniya et al., 2022; Xu et al., 2022; Ambilduke et al., 2023; Sharma and Padha, 2023b; Ondeng et al., 2023). However, because grid features do not provide information at the object level, it is difficult for caption algorithms to produce phrases that are associated with objects. Moreover, the ViT computes global attention

across tiny patches it has divided up from the input image (Dosovitskiy et al., 2020). This is not appropriate where spatially dense prediction is required, such as object detection, because computing expense increases quadratically as image resolution rises. Swin-Transformer (Liu et al., 2022b), which incorporates local attention-supporting techniques like patch reduction and relocated windows, greatly mitigates this problem. While grid-based features are effective in enhancing traditional attention processes to enhance information interaction and relevance analysis, they are usually used for the mapping of internal interactions within one type of visual feature. While grid-based features enhance information interaction, they still lack object-level precision and can become costly at high resolutions.

3.3. Hybrid features

The third stage in visual feature representation involves combining both region attributes and grid attributes as visual inputs to leverage their advantages (Luo et al., 2021; Xian et al., 2022b; Jiang et al., 2022; Wu et al., 2022b; Wang et al., 2023; Parvin et al., 2023; Wu et al., 2022; Nguyen et al., 2022). These hybrid methods fuse region and grid attributes into one visual input and have shown a viable way to address the limitations of grid-based and region-based approaches. The authors in Frome et al. (2013) suggested a double-stream graph convolutional network architecture to handle these two types of characteristics at the region and grid level, respectively. Another strategy, called S2-Transformer, is used in Zeng et al. (2022) to simulate region proposals via clustering on grid features, enriching the visual representation. Additionally, Wu et al. (2022) proposes a two-step interaction approach to deeply integrate region and grid information, enhancing multi-feature expressiveness. The object filter model in Elbedwehy et al. (2022) extracts region features by utilizing Faster-RCNN (Ma et al., 2023) as a detection algorithm for objects and extracts grid features by utilizing the RSTNet backbone (Zhang et al., 2021b). The DLCT model proposed in Luo et al. (2021) creates a graph of geometrical alignment to enhance the precision and quality of by modeling the internal relationships between various forms of features. This method may, however, cause semantic noise while attention is being engaged. Hierarchical representation in Hu et al. (2022a) incorporates instance, region, and grid characteristics and uses LSTM to refine them accordingly, as a technique to improve a deeper understanding of the image. In works such as (Elbedwehy et al., 2022; Nguyen et al., 2022; Luo et al., 2020; Wang et al., 2020), Swin-Transformer is used as a backbone network to extract multi-visual features, which is now standard in many vision tasks.

Although combining object and scene cues offers substantial improvements, challenges remain in terms of computation and fusion complexity. Furthermore, while these hybrid models show promise, future research should focus on reducing computational overhead and refining fusion mechanisms to enable efficient end-to-end image captioning.

4. A taxonomy of image captioning models

With the remarkable success of the Transformer in object identification, NLP, and machine translation tasks, it became a strategy used

by numerous researchers for image captioning tasks (Elbedwehy et al., 2022; Nguyen et al., 2022; Luo et al., 2020; Wang et al., 2020; Tsaniya et al., 2022; Dandwate et al., 2023b). Early approaches utilized the original Transformer decoder over RNNs with minimal modifications to capture long-range dependencies. Over time, significant innovations emerged, particularly in attention mechanisms, pre-training strategies, and feature fusion methods, addressing previous shortcomings such as limited spatial reasoning, semantic coherence, and computational efficiency (Jiang et al., 2020; Guo et al., 2020; Pan et al., 2020; Cornia et al., 2020). Transformer uses self-attention and feedforward network for encoding, which has been proven to be an effective technology in long-term dependency analysis and is utilized in many image caption models (Sharma and Padha, 2023b; Ondeng et al., 2023). Several image captioning models have made use of the standard Transformer as a language decoder without implementing any major architectural changes (Luo et al., 2021; Guo et al., 2020; Herdade et al., 2019; Wang et al., 2021). Several other efforts have been made in the direction of multimodal fusion (Liu et al., 2019; Cornia et al., 2020a; Touvron et al., 2021; Liu et al., 2021a). Simple aggregation processes, such as concatenation, were employed in earlier techniques to merge multimodal sub-networks. Additionally, a few variations have been suggested to enhance the encoding of visual features and language production based on the use of a transformer model for image captioning. Most of the current Transformer-based approaches for image captioning often use single visual characteristics, such as region or grid features. Region attributes are bounding box-based localized image properties that can be discovered with object detectors such as a pre-trained Faster R-CNN, while grid attributes are local characteristics extracted from the full image using the backbone network's high-layer feature map, such as Vision Transformers (ViT). As an alternate paradigm to conventional Transformer designs, diffusion-based generative frameworks have been investigated in recent image captioning advancements. These diffusion models can consistently produce a wider range of caption variations and are more robust to input noise and ambiguities. Despite their natural tendency to produce highly unique descriptions and allow stochastic sampling, diffusion techniques' significant latency and sensitivity to hyperparameter setups make them unsuitable for real-time applications. The following sub-section discusses various approaches that were developed to enhance the model's functionality in image captioning. These approaches are categorized based on the innovation type: Attention Mechanisms, Fusion Strategies, Pre-training Approaches, and Multimodal Large Language Models (MLLMs).

4.1. Approaches based on attention mechanisms

Early image captioning relied on RNN-based attention to focus on salient regions, but Transformers introduced scalable self- and cross-attention to capture long-range dependencies across all visual tokens. Attention mechanisms then evolved significantly, initially employing simple self-attention and later integrating geometric and memory-enhanced strategies to address spatial coherence limitations. For instance, authors in Yang et al. (2019) first adopted a self-attention methodology to encode correlations among features collected using object detectors. Building upon this, the authors in Li et al. (2019) suggested a Transformer-based model that combines a visual encoder of the region's characteristics and a semantical encoder of information from an external tagger. To address earlier self-attention limitations, Attention on Attention (AoA) method proposed in Huang et al. (2019) enhances the visual attributes in which the inquiries and the outcomes of the self-attention are combined, followed by the computation of data and gate vector, and lastly their multiplication. X-linear attention in Pan et al. (2020) improves the representation of the output attended feature by using bilinear pooling approaches. Geometry-aware encoding by Herdade et al. (2019) is a self-attention modification method that considers the spatial interactions among regions, in which scaling attention weights is done by employing an extra geometric weight that

is computed across object pairings. Along similar lines, a normalized and geometry-aware model has been established by Guo et al. (2020), which is also a modification of self-attention that utilizes relative geometry connections among input objects, substantially improving spatial context modeling compared to earlier approaches. M2-Transformer model presented in Cornia et al. (2020) utilizes pre-training Faster RCNN to retrieve region characteristics, relying on the X-Linear attention, further improved contextual feature retention across encoding layers. A spatial graph Transformer presented by He et al. (2020), in which executing attention takes into consideration several sorts of spatial relationships across detections. Advanced models with geometry and memory enhancements significantly enhanced spatial reasoning and contextual recall, in contrast to earlier basic self-attention techniques, although at a higher computational cost. Memory-augmented attention presented by Cornia et al. (2020a) is a Transformer-based architecture in which self-attention is supplemented by a collection of memory vectors at each encoder layer. Another full Transformer approach called Meshed Transformer was introduced by Cornia et al. (2020) uses memory-augmented attention for regional features encoding. In contrast to region-based approaches, current methods such as Zhang et al. (2021b) Ma et al. (2023), Jiang et al. (2020) have started directly applying self-attention to grid features, which drastically reduces computing complexity at the expense of explicit object-level detail. The method brought out by Zhang et al. (2021b) immediately integrates self-attention with grid features while determining their relative geometrical relationships. The use of diffusion-based generative frameworks as an alternate paradigm to conventional Transformer topologies has also been explored in recent image captioning advancements. Diffusion models are computationally and logistically expensive, but they consistently produce a wider range of caption variations and are more robust to input noise and ambiguity. On the other hand, traditional attention-based captioners provide easier deployment and more efficient inference due to their well-established programming libraries and simplified training pipelines. These attention enhancements provide the foundation for MLLMs to perform cross-modal reasoning—applying similar attention blocks jointly over language tokens and image features. Despite the fact that these developments greatly enhanced spatial and contextual comprehension, attention mechanisms continue to face substantial obstacles, which are covered in depth in Section 5. These include:

- **Limited fine-grained interactions:** Self-attention and cross-attention often fail to adequately model subtle spatial relationships between adjacent objects or detailed visual features.
- **Computational overhead:** The complexity and resource demands of global and multi-head attention significantly increase with higher image resolutions.
- **Attention bottleneck:** Transformer models may focus excessively on less informative regions, neglecting crucial visual elements and thus reducing the contextual accuracy of captions.

4.2. Approaches based on pre-training strategies

Prior to Transformer pre-training, captioning systems relied mainly on CNN backbones trained on ImageNet. Pre-training approaches, notably ViT and CLIP, introduced powerful visual embedding techniques to image captioning and enabled end-to-end visual-textual alignment by pre-training on large vision or image-text corpora, addressing the earlier reliance on CNN-based region extraction methods. Transformer structures could be employed directly to image patches (Dosovitskiy et al., 2020; Wu et al., 2020a) to avoid using the convolutional operator. Vision Transformer (ViT) was first introduced in Dosovitskiy et al. (2020) and used for computer vision tasks in which the given image is subdivided into fixed-size chunks, inserted into position-embedded information, and linearly embeddings. The author in Liu et al. (2021a) created the initial convolution-free framework for image

captions in which a standard Transformer decoder is applied to produce captions, and a pre-trained ViT Transformer network is utilized as the encoder. This CaPtion Transformer (CPTR) is a complete-based Transformer model that takes the role of the popular CNN in the encoding phase with a Transformer encoder-decoder structure. Later, Contrastive Language-Image Pre-Training (CLIP) (Radford et al., 2021) and SimVLM (Wang et al., 2021) implemented this visual encoding technique based on grid attributes. CLIP and SimVLM both employ this expanded strategy, significantly enhancing visual-semantic alignment capabilities, where visual encoding in CLIP is trained from scratch on an extensive amount of noisy data. Following that, several approaches for image captioning are made by employing CLIP-based characteristics (Ji et al., 2021; Mokady et al., 2021; Fang et al., 2022; Xian et al., 2022a). While pre-trained transformer models such as ViT and CLIP have significantly advanced captioning, they still encounter notable challenges. Authors in Wang et al. (2022a) proposed a fully Transformer-based architecture called Pure-Transformer, which uses Swin-Transformer (Wu et al., 2022b) instead of Faster-RCNN. A refined encoder and decoder are components of the Pure-Transformer architecture. CTX-M2 model in Kuo and Kira (2022) retrieves extra-textual data in addition to using various visual characteristics from VinVL and CLIP. The AS-Transformer module in Zhang et al. (2022) constructs semantic items for semantic augmentation, followed by an adaptive gating mechanism for balancing semantical and visual input. MLLMs build directly on these pre-trained encoders by coupling them with large language models, enabling richer multimodal understanding and generation. Pre-training strategies have made significant strides, but they are not without drawbacks, which are covered in detail in Section 5. These drawbacks include:

- **Noisy Data:** Models trained on large-scale, noisy datasets may struggle to generalize effectively to more specialized or smaller, domain-specific datasets.
- **Domain-specific Adaptability:** Pre-trained models typically require extensive, computationally expensive fine-tuning to adapt effectively to specialized tasks or distinct domains.
- **Risk of Overfitting:** Extensive pre-training on massive datasets can lead to overfitting, hindering the model's ability to accurately describe unseen or novel visual scenarios.

4.3. Approaches based on fusion strategies

Initial fusion methods simply concatenated region and grid features, but recent Transformer architectures introduced dedicated fusion layers to better integrate heterogeneous visual cues. Most of the current Transformer-based approaches for image captioning often use single visual characteristics, such as region or grid features. Fusion solutions were developed to overcome the drawbacks of single-feature approaches. Early fusion techniques, such as concatenation and summation, suffered from semantic noise and limited feature interaction, motivating advanced methods that leverage the combined advantages of both grid and region visual features more effectively. To make use of the combined benefits of grid and region characteristics, the author in Luo et al. (2021) presented a new hybrid strategy where each type of feature is subjected to two separate self-attention modules, and a cross-attention mechanism locally merges their interactions. This dual-level collaborative Transformer (DLCT) model simulates the interactions between two feature types and relies on local constraints and cross-attention to guide the encoder's semantically based alignment of the region and grid attributes; the encoded data are then directly merged and sent into the Transformer decoder. This approach significantly enhances semantic alignment between visual features compared to basic concatenation methods. Full Transformer-based architecture known as GRIT, presented by Nguyen et al. (2022), employs grid and region-based features in addition to a detector based on DETR to further refine this by parallel cross-attention, effectively improving

multimodal coherence. GET by Ji et al. (2021) included a global vector, which is an average of each feature vector series, to enhance self-attention. The global vector is generated for every layer, and the results are then merged via LSTM to provide an inter-layer representation. A different architecture suggested by Liu et al. (2019) is based on the attention method to match visual phrases retrieved through a concept extractor alongside a grid and present semantically grounded encodings. Expansion Net v2 proposed by Hu et al. (2022a) aids in investigating the potential to enhance the quality of feature refining and subsequently boost the efficacy of the static expansion. Expansion Net design adheres to the conventional encoder-decoder structure based on Swin-Transformer (Liu et al., 2022b). Compared to previous fusion techniques, models like GET (Ji et al., 2021) and Expansion Net v2 (Hu et al., 2022a) improved feature representational power and semantic coherence by dynamically integrating global contextual vectors and transformer-based refinement. However, strategies often faced issues such as semantic noise and ineffective feature interactions. MLLMs extend fusion beyond purely visual features, learning to merge vision and language streams dynamically within a unified Transformer backbone.

While there are still issues with semantic noise and spatial coherence, advanced fusion techniques have enhanced multimodal coherence and feature interaction over straightforward concatenation techniques. The ETA model proposed by Li et al. (2019), for instance, provided a gating mechanism for the cross-attention algorithm, which combines and modulates image region representations with semantical properties obtained by an external tagger to control the flow of visual and linguistic data. Dual-stream structure (DSPLTN) in Jiang et al. (2022) improves multimodal fusion in the decoder. This framework uses adaptive fusing attention to manage interactions across grid characteristics and regions based on linguistic information to enhance word reasoning. DIFNet model in Wu et al. (2022b) develops iterative independent normalized layers to merge segmentation and grid attributes. The dynamic image captioning model in Xian et al. (2022a) constructs a route space with several layers within the encoder based on three separate feature extraction blocks and uses grid characteristics to account for the localization of regional ones. A multimodal Transformer was suggested in Yu et al. (2019) to simultaneously capture internal and inter-modal interactions to integrate visual data from multiple object detectors. The HAN model in Wang et al. (2019a) offers a contextual gating mechanism to flexibly weigh each characteristic's significance on multiple levels and construct a separate multivariate residential property network to merge textual characteristics, regions, and grids at the same time. Pure-Transformer model proposed by Wang et al. (2022a) is an entirely Transformer-based framework that combines the captioning problem into a single step and permits end-to-end training in which Faster-RCNN (Ren et al., 2015) is replaced by Swin-Transformer (Liu et al., 2022b) and the design includes a refined encoder and decoder. According to Zeng et al. (2022), flattening operations may result in the loss of spatial information if they operate directly at grid features. This problem is first addressed by the pseudo-supervised (SP) module and proposed S2-Transformer (Zeng et al., 2022). The detector-free VinCAP framework in Fang et al. (2022) was introduced as an entirely Transformer design that employs grid representations devoid of operations in regional areas. A multimodal Transformer was suggested in Wang et al. (2019a) to simultaneously capture internal and inter-modal interactions to integrate visual data from multiple object detectors. Furthermore, some of the current approaches (Zhang et al., 2022; Kuo and Kira, 2022; Lou et al., 2022; Song et al., 2025; Chen and Li, 2025; Nguyen et al., 2022) focus on raising the decoding capacity for better descriptive creation but fail to include investigating of interactions between attributes while encoding to generate a visual illustration that is stable and complementary. Several features fusion strategies used in earlier studies are described in Fig. 5 and summarized in Table 3.

Although sophisticated fusion techniques improved multimodal coherence, some substantial challenges still exist and are further discussed in Section 5, including:

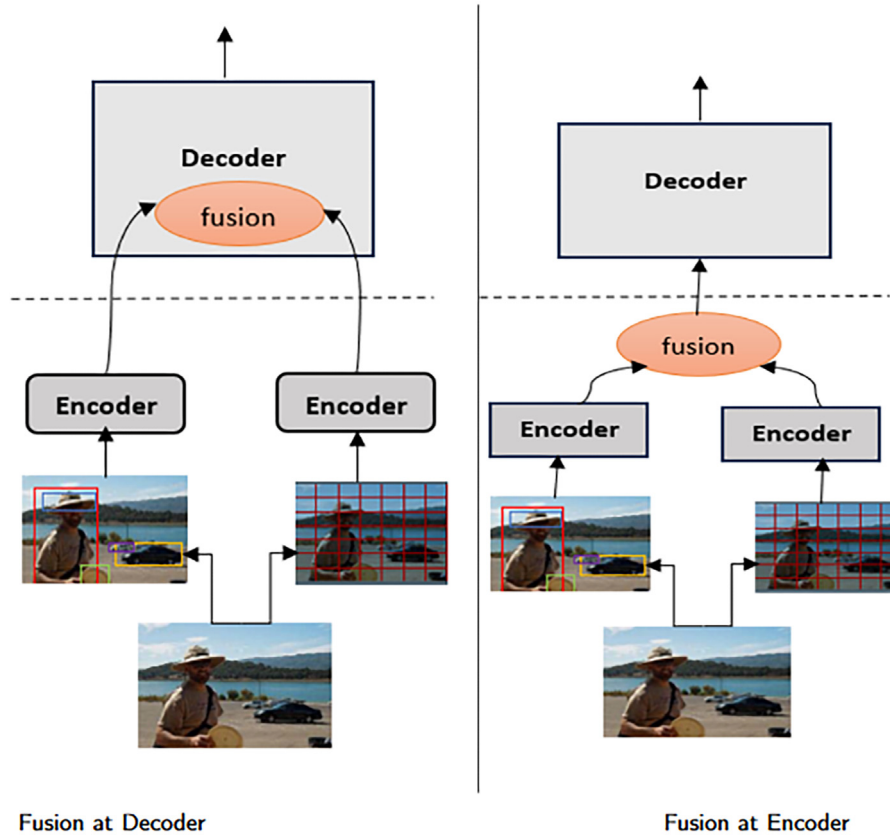


Fig. 5. Comparison of feature fusion strategies.

Table 3
Feature fusion strategies in image captioning models.

Fusion level	Approach	Input	Encoder	Decoder	Fusion method
Encoder	DLCT (Luo et al., 2021)	Region&Grid	Transformer	Transformer	Locally constrained cross attention.
Encoder	Dual-GEE (Xian et al., 2022b)	Region&Grid	Transformer	Transformer	Visual global adaptive attention.
Encoder	DFIENet (Wu et al., 2022b)	Grid&Segment Region	Independent Transformers	Transformer	Iterative independent layer normalization.
Decoder	NADGCN (Wu et al., 2020a)	Region&Grid	Graph-based	RNN-based	Graph scaling and combination.
Decoder	GRIT (Nguyen et al., 2022)	Region&Grid	Independent Transformers	Transformer	Parallel cross attention.
Decoder	DSPLITR (Jiang et al., 2022)	Region&Grid	Independent Transformers	Transformer	Adaptive fusion attention.
Decoder	HAN (Wang et al., 2019a)	Region&Grid and Text	Independent Transformers	LSTM-based	Parallel multivariate residual network.

- **Semantic Noise:** Basic fusion techniques like concatenation or summation can introduce irrelevant or redundant information, diluting the quality of the visual representation and impairing caption accuracy.
- **Ineffective Feature Interaction:** Many existing fusion approaches inadequately model the complex interactions between multiple visual features (e.g., grid and region features), limiting the overall strength of the feature representation.
- **Loss of Spatial Coherence:** Integrating region and grid features often compromises spatial relationships, thereby weakening object-level coherence and reducing the contextual accuracy of generated captions.

4.4. Approaches based on multimodal large language models (MLLMs)

Recent developments have integrated visual and language processing capabilities into Multimodal Large Language Models (MLLMs), such

as LLaVA (Liu et al., 2024) and IDEFICS (Laurençon et al., 2023). By combining pre-trained large language models (LLMs) with robust visual encoders (often derived from CLIP), these models extend traditional text-only LLMs to handle multimodal tasks. These MLLMs, including GPT-4V (Achiam et al., 2023) and Gemini (Team et al., 2023), significantly expand capabilities beyond traditional caption generation, effectively handling complex tasks such as visual dialogue and visual question answering (VQA) (Bucciarelli et al., 2024; Sarto et al., 2025). MLLM architectures typically vary in how visual features integrate with language models. The most common approach in this field is to create multimodal interconnections with multilayer perceptrons (MLP) or linear layers, per the popular LLaVA model family (Bucciarelli et al., 2024). Recent developments by Team et al. (2023) suggested convolutional layers as alternatives to linear layers, while variations like LLaMA-Adapter (Gao et al., 2023) presented alternative processes, including attention modules with zero gating. The Q-Former architecture (Li et al., 2023) efficiently compresses visual

Table 4
Comparative overview of captioning models: Key advances and drawbacks.

Category	Model (Year)	Main Contribution	Drawback
Attention	Object Relation (2019)	Integrates spatial relationships between objects	Sensitive to detector errors.
Attention	Attention on Attention (2019)	Gated double-attention to refine object interactions	High computational requirements.
Transformer	Entangled (2019)	Gated bilateral controller fusing semantic tags with visual features	Requires an external tagger.
Transformer	X-Linear (2020)	Second-order spatial + channel bilinear attention	Memory-intensive.
Transformer	Meshed-Memory (2020)	Multi-layer memory mesh between encoder/decoder	Sensitive to hyperparameters.
Fusion	HAN (2019)	Hierarchical gating across region, grid, and text streams	LSTM overhead.
Fusion	DLCT (2021)	Locally-constrained cross-attention fusion of grid + region features	Potential semantic noise.
Fusion	GRIT (2022)	Parallel cross-attention over region and grid inputs	Dependent on expensive detectors.
Fusion	DSPLITR (2022)	Adaptive fusion attention in the decoder	High inference latency.
Pre-training	OSCAR (2020)	Object tags as anchors for cross-modal pre-training	Tag-quality dependent.
Pre-training	VinVL (2021)	Large-scale fine-tuned object detector for richer regions	Heavy detection overhead.
Pre-training	CLIP (2021)	Contrastive image-text pre-training on 400 M web pairs	Noisy web data.
Pre-training	SimVLM (2022)	Prefix LM for bi-directional + autoregressive generation	Limited fine-grained control.
MLLMs	LLaVA (2023)	Adapter-based vision finetuning of large LLM via CLIP	Hallucination risk; closed weights.
MLLMs	mPLUG-Owl (2023)	Q-Former to compress visual tokens for LLM integration	Significant compute demands.
MLLMs	GPT-4V (2024)	Unified multimodal Transformer for deep reasoning	Proprietary; very high inference cost.

information into fewer trainable tokens. The mPLUG-Owl model in [Ye et al. \(2023\)](#) adds a visual abstractor element. Likewise, Qwen-VL ([Bai et al., 2023](#)) uses learnable queries and single-layer cross-attention to improve the efficiency and compactness of visual data integration. Dense cross-attention blocks integrated into the pre-trained MLLM several layers, often coupled with a Perceiver model ([Jaegle et al., 2021](#)) to minimize visual tokens prior to language model incorporation, have also emerged as notable strategies ([Bucciarelli et al., 2024](#)). Despite their advanced capabilities, MLLMs introduce distinct challenges and limitations (discussed in Section 5), such as:

- **Evaluation Complexity:** Traditional captioning metrics may not effectively evaluate the richer, nuanced outputs of MLLMs.
- **Hallucination and Grounding:** Ensuring the factual correctness and grounding of MLLM-generated captions remains challenging, requiring novel evaluation and validation approaches.
- **Computational Overhead:** The significantly higher computational requirements of MLLMs pose efficiency concerns for practical implementation.

To summarize the critical challenges across image captioning models, [Table 4](#) provides a comparative overview of key image captioning categories, highlighting their innovations and challenges. The shift from basic attention-based methods to complex multimodal big language models demonstrates the rapid evolution and the increasing complexity of the image captioning models. Further research is still needed to address these challenges, positioning MLLMs as the next frontier in Transformer-based multimodal understanding.

5. Limitations of image captioning models

This section provides detailed insights by expanding on the obstacles briefly mentioned, along with the related Transformer-based approaches described in Section 4.

5.1. Attention mechanisms limitations

Most of the current Transformer-based approaches for image captioning often use single visual characteristics, such as region or grid features, and fail to consider the spatial interactions between multiple visual features ([Lou et al., 2022](#)). The success of higher-level trained grid features, the development of self-attention visual encoders, and enormous-scale multimodal systems are some elements that have fueled the ongoing discussion about the optimal feature model for image captioning ([Stefanini et al., 2022](#)). For captioning image models based on Transformers, the image is initially expressed as a collection of regional attributes. The interaction between local regions is then modeled using self-attention and a scalar (such as attention weight), all of which are utilized to enhance the visual representation of local regions. Nevertheless, numerous ground-breaking studies ([Abdal Hafeth and Kollias, 2024](#); [Elbedwehy et al., 2022](#); [Frome et al., 2013](#)) have shown that, regardless of the distinct advantage of self-attention, regional representations continue to fail to possess image-level attributes. As a result, the decoder cannot receive enough visual cues from these region features, which might result in a major issue with an object being absent. Additionally, concentrating on specific regions might lead to semantic isolation ([Zhang et al., 2023b](#)). Features at the regional level have multiple issues. As the regions do not include those gaps between objects, they first fail to communicate contextual details such as the correlation between the objects. Second, the possibility of incorrect detection and essential objects is ignored. Finally, the calculation of the region-level characteristics is computationally expensive, particularly when utilizing high-performance CNN-based detectors such as Faster R-CNN. This makes it challenging to train the complete model end-to-end ([Nguyen et al., 2022](#)). The aforementioned incident highlights the limitations of Transformer methods based on regional characteristics in terms of capturing cross-region correlations ([Abdal Hafeth and Kollias, 2024](#); [Frome et al., 2013](#)). Some approaches are starting to reconsider the use of grid features to make up for the aforementioned restrictions.

However, Transformer-like networks do not provide the perception of entire objects. Specifically, the flattening process in Transformer invariably damages the local link between grid features, whereas a single object might be partitioned into numerous nearby grids (Frome et al., 2013). Grid-based feature extraction preserves the original structure and considers only the local spatial information of the pixel. It can reduce the possibility of erroneous object detection. This is because region features are derived based on splitting the image into tiny local sections, while grid features are derived based on a high-layer map of attributes from the full image [1-5]. On the other hand, some grid-based approaches utilize a Vision Transformer (ViT) as based instead of the standard Transformer. However, ViT calculates global attention across tiny patches that are created from an input image. As a result, the computational burden grows quadratically coupled with image resolution (Wu et al., 2022b), making it unsuitable for spatial dense prediction, such as identifying objects, which is required for image captions. Previous studies have demonstrated that grid characteristics outperform region characteristics in terms of time cost and performance (Zhang et al., 2021b; Abdal Hafeth and Kollias, 2024; Frome et al., 2013; Zeng et al., 2022). However, because grid features do not include object-level details, it is difficult for caption models to produce words that are associated with objects. While grid-based and regional-based methods enhance traditional attention processes and efficiently support information interaction and relevance analysis, they are usually used to describe internal relationships within a single class of visual features.

5.2. Pre-training strategies limitations

Many image captioning Transformer models use pre-trained networks like ViT or CLIP to enhance visual feature extraction. These models are trained on large datasets and then fine-tuned for captioning tasks. However, these pre-training strategies introduce both benefits and challenges. ViT (Dosovitskiy et al., 2020) and CLIP (Radford et al., 2021) have been leveraged for visual encoding in several Transformer-based models. ViT divides images into fixed-size chunks, while CLIP uses a large-scale, noisy dataset for training. While pre-training was used by models like OSCAR, Unified VLP, VinVL, and VIVO, VIVO concentrated on a distinct type of pre-training that relied on developing a visual terminology and pre-training without the inclusion of full image captions. Although these approaches provide robust visual embeddings, they often suffer from the challenges of domain adaptation and data inefficiencies.

The limitations of Transformer approaches can also be categorized into three main parts:

(1) In the encoding phase, the majority of current Transformer-based models primarily rely on single features like regional or grid characteristics as their standard visual information representation. The locally noticeable parts within the image can be recognized at the object level using region characteristics. However, modeling a single feature, specifically regional characteristics, can result in several limitations: First, regional features do not provide accurate global image background information. Second, erroneous object detection poses a concern, leading to issues such as the misidentification of relationships between items and the absence of tiny objects. Third, a high computational cost is associated with calculating region features. This is particularly true when employing high-performance visual-language pre-trained CNN-based detectors like Faster R-CNN (Zhang et al., 2023b). To address this, other approaches revisit the use of grid features, which are intended to extract all patch information over the full image (Ma et al., 2023; Wang et al., 2022a). Although these methods have demonstrated better efficiency and lower computational costs compared to regional-based ones, Transformer architectures are not suited to recognizing complete objects. This is because the entire object could be segmented into multiple adjacent grids, and the flattened operation of the Transformer could damage the local relationships,

which consequently causes semantic information damage and makes reasoning more difficult (Yang et al., 2023; Wu et al., 2022). The grid and regional methods have difficulty managing interactions between several visual features because they frequently depict internal connections in a single feature while ignoring the advantageous effects of using multiple features to obtain more superior feature representation during the encoding stage, while only a few recent studies (Xian et al., 2022b; Nguyen et al., 2022; Ren et al., 2024; Lou et al., 2022; Zhang et al., 2022) have considered the fusion of multiple features.

(2) In the decoding phase, the aforementioned drawbacks lead to a restricted visual information flow, which forces the language decoder module to heavily rely on partially formed captions to guarantee the generation process and create the remaining words. In this case, the decoder does not obtain sufficient semantic information, especially in the initial stage step, which leads to the creation of erroneous words in the subsequent time step, which ultimately makes the generated descriptions unrelated to the visual content and reduces the quality of captions (Jiang et al., 2022; Wang et al., 2022a; Lou et al., 2022). The recent research directions focused on the fusion of grid and regional characteristics as a way to determine the ideal visual information representation for image captioning; however, the key lies in how to integrate multi-feature information effectively. Current multi-feature approaches still fail to effectively integrate the relationship between different visual features, and the optimal method remains undefined.

(3) Fusion: Transformer-based models restrict complex fusion interaction by often assigning weights of attention to the total candidate vectors, assuming that these vectors are all related during the encoding process. Additionally, during the decoding process, they only use one cross-attention layer for multimodal fusion, omitting hierarchical interaction and intricate relationships involving the query and key vectors (visual and semantic aspects).

5.3. Fusion strategies limitations

According to the earlier study, each feature has its own benefits and drawbacks of its own. Although regional features lack background and relationship information, they do provide explicit object information. Grid features, on the other hand, contain all information simultaneously while allowing for the division of an object into several grids. Recent image captioning research (Luo et al., 2021; Xian et al., 2022b; Lou et al., 2022; Zhu et al., 2022; Dandwate et al., 2023b) has begun to concentrate on merging grid and regional components to boost the representation of visual data. The adoption of the combined region and grid attributes, such as NADGCN (Kuo and Kira, 2022) and DSPLTN (Lou et al., 2022) is a simple way to benefit from both characteristics. However, the traditional fusion methods employed in these approaches may create semantic noise that interferes with model prediction, where an effective fusion technique is still needed. Since much of the semantic content is being deteriorated, reasoning becomes more challenging. Many methods have also been developed to leverage semantic data gathered from separate object detectors to improve the Transformer model for higher performance. However, irrelevant items in the semantic data obtained by offline detectors might introduce noise during the decoding operation (Zhang et al., 2022). Additionally, image captioning depends on the efficient fusion of semantic data with visual elements during the decoding process. The majority of current Transformer-based models simply employ cross attention over created words and visual characteristics before producing the next word, using image region and grid attributes instead of global characteristics (Yang et al., 2023). Most of the current approaches explicitly model the intricate relationships that exist between the local region and the entire image, which results in suboptimal representations of the image. Furthermore, applying attention to several levels of visual characteristics independently before merging attention outcomes leads to subpar intermodal fusion.

Table 5
High-level comparison of single- vs. dual-stream transformer architectures for image captioning.

Architecture	Workflow	Characteristics & Constraints
Single-Stream	Combine tokens into a single Transformer encoder-decoder	Pros: Shares weights across modalities for parameter-efficiency; simpler implementation. Cons: Early fusion might prevent subtle interactions and reduce cross-modal, fine-grained inputs.
Dual-Stream	Separate encoders, cross-attention fusion, joint decoding	Pros: Dedicated pathways allow richer, modality-specific feature processing and stronger semantic grounding. Cons: The number of parameters, memory use, and delay are increased by more encoder blocks and cross-attention layers.

The two primary issues with current fusion transformer methods based on multi-visual features are their inability to carry out intricate multimodal reasoning during decoding and their incapacity to incorporate the benefits of various visual characteristics while encoding. Some studies used the direct concatenation strategy, but they failed to consider the link that is present across region features and grid features, where the feature fusion is carried out only in the encoding level. After that, it is simply concatenated or fused into a single structure and fed into the decoder. Other studies consider a straightforward feed strategy of image region and grid attributes over cross-attention, where feature fusion is carried out only at the decoding level. However, some studies have shown that directly employing both visual characteristics can result in semantic disturbances through basic cross-attention, including concentrating on irrelevant regions and obstructing model inference (Rennie et al., 2017; Zhang et al., 2023b). In contrast to the encoding phase, which lacks feature interaction, the decoding step involves multimodal data fusion. To resolve this problem, recent studies have demonstrated that additional visual information may be efficiently extracted using local modeling of visual characteristics (Frome et al., 2013; Ren et al., 2024). Two general fusion paradigms have been developed to solve these issues: dual-stream models that preserve distinct modality pathways prior to cross-attending, and single-stream models that execute early, unified fusion of visual and textual elements. Efficiency, expressive capacity, and implementation complexity are all balanced differently in each technique. Single stream and dual stream are compared at a high level in Table 5, which highlights their unique benefits and drawbacks.

5.4. Multimodal large language limitations

MLLMs demonstrate outstanding zero-shot capabilities, but are challenging to successfully fine-tune to fit certain domain benchmarks and captioning styles (Bucciarelli et al., 2024). Even with these quick developments, there is still a lack of performance studies targeted at image captioning benchmarks (Sarto et al., 2025). Since many MLLMs include image description into their more general multimodal reasoning skills, limited models are fully trained and evaluated against conventional captioning datasets like MS-COCO. According to Bucciarelli et al. (2024), conventional MLLMs frequently generate captions that deviate from the concise, object-centric, and grammatically sound style typically expected in standard image captioning datasets. Although fine-tuning can enhance performance on specific benchmarks, it may compromise the model's generalization ability, leading to diminished robustness across diverse image domains. Efficiency and scalability are also major concerns due to the high computational demands of training and implementing multimodal large language models (MLLMs). Additionally, Hallucination is a key issue in image captioning research, as models typically incorporate incorrect or manufactured features into their outputs. MLLMs cause complicated hallucinations, including skewed relationships and the invention of entire scene elements (Sarto et al., 2025). Hallucinations in MLLMs, in contrast to typical captioning errors, frequently involve subtle semantic distortions, like incorrect object relations or completely made-up elements. Addressing these sophisticated hallucinations requires benchmarks and mitigation

strategies that go beyond simple token-level checks, focusing instead on semantic consistency and groundedness to ensure that generated captions faithfully reflect the visual content.

6. Discussion and analysis

This section compares and analyzes various cutting-edge image captioning models based on their performance with the MS-COCO dataset (Lin et al., 2014) using Karpathy splits (Karpathy and Fei-Fei, 2015). Comparing the performance of Karpathy splits is crucial because not all methodologies are available on the Web server. Table 6 summarizes numerous approaches based on their experimental outcomes using the standard MS-COCO dataset and widely recognized metrics: B-4 stands for BLEU-4, S for SPICE, M for METEOR, R for ROUGE-L, and C for CIDEr. VLP and non-VLP models are referred to as yes or no.

6.1. Feature extraction impact

One of the most critical aspects of transformer-based image captioning models is the choice of feature extractor. Faster R-CNN has been a standard for extracting visual features from images for a long time, and it is a key component of traditional models like Li et al. (2019), Ji et al. (2021), Pan et al. (2020). In contrast, image captioning tasks have seen a notable improvement in performance since the rise of transformer-based feature extractors (Wang et al., 2022a; Lu et al., 2023; Nguyen et al., 2022). Models such as Yu et al. (2022), for instance, use a Vision Transformer (ViT) to directly extract features from image patches. Compared to Zhou et al. (2020), Wang et al. (2022c) reports higher scores of BLEU-4 (39.5 vs. 40.1) and CIDEr (143.6 vs. 131.6), indicating the superiority of patch-based extractors over Faster R-CNN. Some patch-based models, such as Pure-Transformer, CoCa, and GIT, achieve an average BLEU-4 of 41.3, representing an absolute improvement of 2.6 points over R-CNN-based baselines. Even though OSCAR's object detector was Faster R-CNN, VinVL's visual encoder improved its performance by encoding significantly more visual information and object tags. Patch-based encoders outperform traditional object-level detectors in capturing global image context. However, for lengthy input sequences, transformer-based systems are incredibly slow. A variation of the original vision transformers, Swin Transformer (Liu et al., 2022b1), offers a solution to the transformers' quadratic complexity, which is crucial for processing higher-quality images. Faster R-CNN excels at object-level detection, but it has difficulty comprehending global details; in contrast, Vision Transformers are better at capturing spatial relationships.

6.2. Optimization technique comparison

Optimization techniques play a key role in the success of transformer-based image captioning models. The vast majority of current models optimize performance using cross-entropy loss during training. However, more recent methods have turned to Reinforcement Learning (RL) to fine-tune models after initial training. The Self-Critical Sequence Training (SCST) optimization, developed by Rennie et al. (2017), is a more contemporary method that is further

Table 6

Performance of numerous Transformer-based models on the MS-COCO dataset with Karpathy test split.

Author & Year	Method	Feature Type	Feature Module	Visual Module	Language Module	VLP	B-4	M	R	C	S
Zhu et al. (2018)	Stacked Attention	Region& Text	ResNet-101	CNN	Transformer	No	33.3	–	54.8	108.1	–
Herdade et al. (2019)	Object-Relation	Region	Faster R-CNN& ResNet101	Self-Attention	Transformer	No	38.6	28.7	58.4	128.3	22.6
Huang et al. (2019)	AoA	Region	UpDown-R101	Self-Attention	LSTM	No	38.9	29.2	58.8	129.8	22.4
Yu et al. (2019)	Multimodal Transformer	Region	ResNet101, ResNet152, ResNeXt101	Self-Attention	Transformer	Yes	–	29.5	–	134.1	–
Li et al. (2019)	Entangled Transformer (ETA)	Region& Text	VGG-16	Self-Attention	Transformer	No	39.9	28.9	59.0	127.6	22.6
Wang et al. (2019a)	HAN	Region& Grid& Text	Faster R-CNN& ResNet101	Self-Attention	LSTM-based	No	37.6	27.8	58.1	121.7	21.5
Cornia et al. (2020)	M2-Transformer	Region	ResNet-101	Memory-Augmented Attention	Transformer	No	39.1	29.2	58.6	131.2	22.6
Pan et al. (2020)	X-Transformer	Region	Faster R-CNN& ResNet-101	X-linear Attention	Transformer	No	39.7	29.5	59.1	132.8	23.4
Guo et al. (2020)	NG-SAN	Region	Bottom-Up	Self-Attention	Transformer	No	39.9	29.3	59.2	132.1	23.3
He et al. (2020)	Trans-KG	Region	Bottom-Up	Self-Attention	Transformer& Knowledge Graph	No	39.5	29.1	59.0	130.8	22.8
Zhou et al. (2020)	Unified VLP	Region	Faster R-CNN	Self-Attention	Transformer	Yes	39.5	29.3	–	129.3	23.2
Li et al. (2020)	OSCAR	Region& Text	Faster R-CNN	Transformer Object Tags	BERT base	Yes	41.7	30.6	–	140.0	24.5
Wu et al. (2020a)	NADGCN	Region& Grid	ResNet101& GCN	Graph-based Encoder	RNN-based decoder	Yes	38.3	28.5	58.2	126.4	21.8
Luo et al. (2021)	DLCT	Region& Grid	Up-Down-R101& Jiang-X101	Self-Attention	Transformer	No	39.8	29.5	59.1	133.8	23.0
Ji et al. (2021)	GET	Region	Faster R-CNN	Global-enhanced Attention	Global Adaptive-Transformer	No	39.5	29.3	58.9	131.6	22.8
Mokady et al. (2021)	ClipCap-Transformer	Grid	CLIP	Transformer based Mapping Network	GPT2	Yes	33.5	27.6	55.9	112.8	21.0
Zhang et al. (2021a)	VinVL	Region	ResNeXt-152 C4 layer	Transformer	Transformer-based Fusion Model	Yes	41.0	31.1	–	140.9	25.2
Zhang et al. (2021b)	RSTNet	Grid& Text	ResNext152 &Jiang-X152	Augmented Module &Pre-trained BERT	Relationship-Sensitive Transformer	Yes	40.1	29.8	59.5	135.6	23.3
Wang et al. (2021)	SimVLM	Grid	ViT/ CoANet	Self-Attention	Transformer	Yes	40.6	33.7	–	143.3	25.4
Fan et al. (2021)	TCIC	Region& Text	Faster R-CNN	Self-Attention	Transformer with Theme Nodes	No	39.7	29.2	58.6	132.9	22.4
Hu et al. (2021)	VIVO	Region	UpDown	Self-Attention	Multilayer Transformer	Yes	34.9	28.4	–	119.8	21.7
Hu et al. (2022b)	LEMON	Region& Text	Faster R-CNN	Multilayer Transformer	Multilayer Transformer	Yes	42.6	31.4	–	145.5	25.5
Wang et al. (2022b)	OFA	Grid& Text	ResNet152	Self-Attention with post-attention layer	Transformer	Yes	43.6	32.2	–	150.7	26.2
Yu et al. (2022)	CoCa	Region& Text	ViT/ ConvNet	Transformer	Decoupled Transformer	Yes	40.9	33.9	–	143.6	24.7
Wang et al. (2022c)	GIT	Grid	CoSwin Yuan et al. (2021)	Contrastive pre-trained Model	Transformer	Yes	44.1	32.2	–	138.1	26.3
Li et al. (2022)	COS-Net	Grid& Text	CLIP	Self-Attention	Semantic-Ranker& Transformer	No	42.0	30.6	60.6	141.1	24.6
Wang et al. (2022a)	Pure-Transformer	Grid	Swin-Transformer	Refining-Transformer Encoder	Transformer with Pre-Fusion Module	No	40.9	30.2	60.1	138.2	24.2

(continued on next page)

Table 6 (continued).

Jiang et al. (2022)	DSPLTN	Region& Grid	Up-Down-R101& Jiang-X101	Transformer	Transformer	No	39.4	29.5	59.0	133.3	23.1
Xian et al. (2022b)	Dual-GEE	Region& Grid	Faster R-CNN& ResNet-152	Transformer	Transformer with textual global adaptive attention	No	40.3	29.2	59.4	132.4	23.3
Wu et al. (2022b)	DIFNet	Grid& Segment grid	Jiang-X101& UPSNet-R101	Transformer	Transformer& Independent Layer Normalization	No	40.0	29.7	59.4	136.2	23.2
Nguyen et al. (2022)	GRIT	Region& Grid	Swin-Transformer	Independent Transformer	Parallel Cross-Attention	Yes	41.9	30.5	60.5	142.2	24.2
Huang and Li (2022)	SGT	–	–	–	–	No	38.7	28.9	58.5	129.9	–
Kuo and Kira (2023)	HAAV	Region& Grid& Text	Bottom-Up& CLIP ViT-B/32	Independent-Transformer	Hierarchical-Transformer	No	41.0	30.2	–	141.5	23.9
Zhang et al. (2023a)	IMFRE-Transformer	Region& Segment Region	Faster R-CNN& ResNet101	MFF-Transformer	MLFF-Transformer	No	36.4	28.3	57.1	117.1	21.6
Hafeth et al. (2023)	SGA	Region	Faster R-CNN	Transformer	LSTM	No	36.0	27.6	57.7	120.9	–
Lu et al. (2023)	FMT	Region	UpDown/Faster R-CNN	Full-Memory Transformer	Transformer	No	39.3	28.4	58.5	126.5	22.3
Zheng et al. (2023)	TRET	Region& Grid	Faster R-CNN& Graph Convolution Network	Relationship Enhanced Transformer	Multi-layer Attention	Yes	37.5	28.2	57.8	126.1	–
Li et al. (2024)	VG-Cap	Grid	ResNeXt-101	Enhanced-adaptive Graph Relation Attention	Semantic Fusion Cross Attention-Transformer	No	42.3	30.3	60.6	136.6	24.1
Abdal Hafeth and Kollias (2024)	OSLT	Region	Faster R-CNN& ResNet-101	ConceptNet Knowledge	Transformer	No	37.7	28.9	58.5	132.0	–
Mundu et al. (2024)	ETransCap	Region	ConvNext	Transformer	Transformer with Efficient Attention	Yes	38.0	28.4	58.1	127.8	–
Basak et al. (2024)	TICOD	Region	Swin-Transformer	Transformer	GPT-2 Transformer	Yes	35.3	28.3	56.7	115.3	21.1
Bui et al. (2024)	TMDNet	Grid	ResNeXt-152/VinVL	Depth-Grid Attention	Transformer	Yes	40.4	29.9	59.7	137.0	23.4
Chen and Li (2025)	MMGAT	Region& Grid& Text	Faster R-CNN & Swin-Transformer& CLIP-T	Cross-modal Aggregation	Transformer	Yes	42.5	31.1	60.8	144.6	24.6
Xu et al. (2024)	DAMAT	Grid	Faster RCNN-C5	Distance-aware Self-Attention	Multilayer Aggregated Network& Transformer	Yes	39.3	29.5	58.9	133.2	23.1
Song et al. (2025)	TRT-AP	Grid	ViT	Two-modal Fusion	Transformer	No	41.5	30.5	61.1	142.1	24.7
Ren et al. (2024)	–	Region& Grid	Faster R-CNN& ResNet-101	Region-wise Attention& Align-Cross Attention	Transformer	No	39.4	29.3	59.1	133.1	23.9

employed by a significant number of models and has established its role as a standard optimization method for image captioning. Reinforcement learning is applied in SCST optimization to explicitly optimize for the CIDEr evaluation measure, which is an important metric used to compare various models and present outcomes on the MS-COCO captioning. Following initial training with cross-entropy optimization, SCST is employed for fine-tuning. For instance, models such as (Herdade et al., 2019) rely solely on traditional cross-entropy loss, while they perform well, they do not reach the levels of accuracy achieved by models using SCST or RL-based techniques like (Li et al., 2020; Wu et al., 2020a). This shows that SCST, which optimizes for CIDEr, a metric known to better capture the relevance of captions to the ground truth, is crucial for improving caption quality.

6.3. Semantic accuracy challenges

Despite achieving high scores on common metrics, many transformer-based models struggle with semantic accuracy in image captioning. Although metrics such as CIDEr, BLEU, and METEOR offer a numerical performance assessment, they are not always correlated with the quality or relevance of captions. For example, models like Mokady et al. (2021) achieve high CIDEr using a grid-based feature extraction method with CLIP, but still fail to produce semantically accurate captions in some cases. While they generate high-quality captions in terms of linguistic fluency, the captions may lack true semantic alignment with the visual content, leading to confusion or irrelevant descriptions in certain contexts. Conversely, models like Luo et al. (2021), which integrate region and grid features, demonstrate better

semantic alignment. Although the CIDEr score is slightly lower than that of CLIP-based models, the captions generated are more contextually accurate, highlighting the importance of combining multiple visual features (region and text) to capture both detailed object-level information and the global image context.

6.4. Real-world consistency analysis

The models analyzed in Table 6 demonstrate how optimization techniques and architectural decisions influence image captioning performance. When it comes to collecting global context, Vision Transformer-based encoders have outperformed Faster R-CNN backbones, eliminating the need for explicit object detectors. Although most systems still use cross-entropy optimization and SCST fine-tuning for CIDEr gains, genuine semantic integrity is still unresolved. Despite the success of Transformer, current Transformer frameworks still have significant limitations in generating semantically accurate, adequate, and natural textual-based descriptions of full image content (Li et al., 2020; Hu et al., 2021; Zhou et al., 2020; Zhang et al., 2021a; Mokady et al., 2021; Wang et al., 2021; Hu et al., 2020; Jia et al., 2021; Chen et al., 2022; Zhu et al., 2018; Li et al., 2022; Wang et al., 2022b; Yu et al., 2022). Moreover, despite advances in these areas, achieving semantic accuracy remains a significant challenge, as models often struggle to generate captions that fully align with visual content. Since each image might vary in complexity and context, there is still no clear-cut boundary for what makes a great caption (Salgotra et al., 2024; Li et al., 2024; Sharma and Padha, 2023a; Ming et al., 2022; Stefanini et al., 2022; Ghandi et al., 2022; Kavitha and Karpagam, 2022; Osman et al., 2023). Although Transformer and LLM-based captioners have seen remarkable improvements, thorough human tests continue to show gaps in real-world performance. Closed-weight models such as GPT-4o reach human-level agreement on complex captioning benchmarks, according to CapArena's extensive analysis, while top open-source systems fail short in terms of human consistency (Cheng et al., 2025). Furthermore, while modern Vision-Language approaches achieve a good correlation with human preferences, standard automatic measures like BLEU and CIDEr consistently underreport this alignment (Cheng et al., 2025). In addition to these overall scores, CapArena also determines some primary causes of failure that continue to affect state-of-the-art systems: (1) misrepresenting uncommon or out-of-distribution scenes; (2) overlooking subtle yet crucial visual details; (3) struggling with scene-spanning knowledge relationships (Cheng et al., 2025). The following section outlines promising future directions to address these challenges.

7. Future directions and prospective challenges

This paper investigates the potential of Transformer models in image captioning and highlights their superior performance compared to earlier approaches. Despite clear progress, several unresolved challenges remain, which create ample room for further research. This section discusses promising future directions in image captioning. The evidence reported in the previous section identifies the following development paths for image captioning.

7.1. Enhancing visual feature representation

- Many captioning methods have largely overlooked the feature extractor, instead focusing on the encoder-decoder components. Following the bottom-up approach (Anderson et al., 2018) and Faster R-CNN (Ren et al., 2015), which became the industry standard for object detection, recent advances in object detectors have led to improved performance. The VinVL model, for example, enhances the feature extractor by improving object detection and training on larger datasets. Future work is expected to continue refining these detectors to provide richer features to captioning pipelines.

- Although most image captioning systems commonly use Transformers for encoding and decoding; however, many still rely on CNNs for visual feature extraction. Recent studies show that pure Transformer encoders (e.g., Vision Transformers) can rival CNNs, leading to end-to-end architectures that eliminate the need for CNNs and explicit object detection.
- The advent of fully Transformer-based detectors, such as DETR and its many enhancements, has simplified end-to-end vision modeling and improved efficiency. The visual component of captioning models, as well as other computer vision tasks, could benefit substantially from further research and advancement in these domains (Yuan et al., 2021).

7.2. Improving multi-modal coherence

- Image captioning is one of the many tasks that have demonstrated the effectiveness of models integrating multiple modalities, particularly those combining vision and language. A key challenge in this domain is to identify more efficient methods to enhance cross-modal representations across these two modalities. Enhancing the multimodal alignment is greatly aided by utilizing extensive text-image-pair databases for vision-language pre-training employing contrastive and masked language modeling losses. However, further research is still needed to develop more efficient approaches for aligning visual features with linguistic elements, which could lead to more accurate and comprehensible image captions.
- Image captioning is undergoing rapid advancement in both object detection and description generation. The ability of models to extract relevant multimodal segments enhances semantic understanding and improves representation accuracy. However, a major limitation remains the scarcity of high-quality datasets, especially in critical domains such as medical image captioning and visual assistance. If inconsistent sampling is not addressed during training, models may produce unreliable outputs. Thus, developing methods that learn effectively from limited, heterogeneous samples and curating high-quality benchmarks for specialized domains are other critical research directions (Wu et al., 2022; Zhou et al., 2022).
- Transformer architecture was established in the language domain and is well-suited for NLP problems, despite its further application in computer vision. Therefore, further investigation is still needed to develop vision-tailored Transformer architectures that are more suitable for image captioning and other applications involving vision.

7.3. Addressing training complexity and efficiency

- Training Transformer-based image captioning models remains computationally demanding due to the complexity of multi-head attention mechanisms and the need for extensive hyperparameter tuning. This process often involves multiple stages and typically requires high-end GPUs, posing a significant barrier for researchers with limited hardware resources. To reduce training time and computational requirements, future research should explore strategies like model distillation, parameter-efficient tuning methods, and lightweight Transformer variants to create more accessible and scalable captioning systems.
- Models trained on a specific language or domain often fail to generalize effectively to new linguistic or contextual settings without requiring full retraining. This limitation highlights the need for accurate and efficient multimodal transfer learning strategies to enable seamless cross-domain and cross-lingual knowledge transfer in image captioning. Despite the growing importance of this capability, progress in developing robust multi-domain adaptation techniques for image captioning remains limited, presenting a significant opportunity for future research.

- Although Transformers scale effectively with large datasets, their lower inductive bias compared to CNNs and LSTMs increases reliance on vast amounts of data and computational resources (Stefanini et al., 2022; Sharma and Padha, 2023a; Abdal Hafeth and Kollias, 2024). To make large-scale multimodal models more accessible and efficient, future work should focus on incorporating vision-language specific inductive biases and designing more efficient training objectives. Most current large models depend on either novel training objectives or modifications to the core training pipeline, yet there remains considerable potential in developing Transformer-based architectures that balance scale with cost-effectiveness.
- Enhancing self-supervised pretraining for vision Transformers remains a promising research direction, particularly in combining unsupervised and supervised signals to learn more robust visual representations. By reducing reliance on large labeled datasets, such approaches can improve model generalization and scalability. Future research might focus on developing more efficient self-supervised objectives and architectures to bridge the gap with extensive supervised pretraining and allow for wider application in fields with limited data.

7.4. Handling rare objects and caption consistency

- Effectively handling long-tail objects, accurately identifying object relationships, and ensuring consistency across generated descriptions remain significant challenges in image captioning. These limitations often lead to hallucinations or misclassifications, especially in cases involving rare objects or underrepresented scene elements in the training data. Future research should focus on strategies that improve model robustness and generalization in such scenarios, such as incorporating external knowledge, leveraging data augmentation, or developing specialized modules for rare object recognition and relational reasoning.
- Retrieval-augmented captioning techniques represent a promising direction for improving accuracy in handling rare or unseen objects. By retrieving relevant captions or contextual information from large external databases, these methods can provide more grounded and informative context during generation, reducing hallucinations and enhancing the model's ability to describe infrequent or novel visual concepts. Future research should explore how to effectively integrate retrieval mechanisms with captioning models to balance both accuracy and efficiency.

7.5. Exploring image captioning evaluation in the age of MLLMs

- With their enhanced reasoning and generative capabilities, MLLMs represent a natural evolution beyond traditional image captioning systems. MLLMs sit at the forefront of vision-language research, handling complex tasks such as long-form visual description generation and Visual Question Answering (VQA) in addition to captioning, which has significantly shaped the field's direction. The faithfulness and grounding of MLLM-generated content, however, continue to be a major challenge, especially when dealing with hallucinations, in which the model generates information that is not present in the visual input. Future studies require establishing varied and precise benchmarks that include specific test cases (such as attribute confusions and nonexistent objects) along with specialized evaluation metrics capable of identifying and measuring these high-level semantic errors to overcome these limitations.
- Existing image captioning metrics (e.g., BLEU, METEOR, CIDEr) may not be sufficient to evaluate the richer outputs produced by MLLMs, which often go beyond standard caption generation into more complex outputs (Sarto et al., 2025). New or modified evaluation frameworks that go beyond surface-level lexical

matching and more closely reflect human-like understanding are desperately needed to accurately capture the depth, coherence, and reasoning abilities of these models.

- MLLMs have significant computational infrastructure requirements during both training and deployment, posing challenges for efficient implementation, scalability, and efficiency. Future research should explore lightweight evaluation pipelines and scalable, model-agnostic scoring frameworks that maintain robustness while reducing resource demands.

Beyond these directions, by introducing new transformer structures, there is still scope for development within transformer-based models, which might enhance the overall accuracy of the resulting description. Future research may yield Transformer variant that are not only more effective but also better suited for everyday applications (Sakirin and Said, 2025). Moreover, progress in image captioning has the potential to extend beyond static images, contributing to related domains such as video understanding and broader multimodal tasks, thereby opening new avenues for cross-media semantic interpretation.

8. Conclusion

This study reviews advancements in image captioning models with a particular focus on transformer-based approaches, which are the building blocks of current state-of-the-art models. While traditional frameworks have been sufficiently reviewed in previous studies, this paper offers a taxonomy of transformer-based methodologies and discusses their characteristics and qualities. The paper also analyzes their common shortcomings, evaluates the performance of the discussed approaches based on experimental data, and demonstrates the obstacles and future possibilities in image captioning. Transformers have enabled significant advancements in generating descriptive captions, unlocking new research avenues in the field. Despite the great potential of transformer-based image captioning, there are several crucial challenges and obstacles for which few solutions have been proposed. One of the key areas of focus in transformer-based image captioning is to explore more efficient and effective methodologies to enhance the quality of generated captions. Transformer-based approaches remain prevalent in vision and language frameworks, and their application in image captioning is a growing field of research. Furthermore, the rise of Multimodal Large Language Models (MLLMs) is redefining the landscape of image captioning by integrating multimodal data more seamlessly. As the field advances, addressing challenges such as hallucinations in generated content and improving multimodal coherence will be essential. The future of image captioning is promising, with broad application potential; ongoing research will be vital for overcoming these obstacles and advancing the discipline.

CRedit authorship contribution statement

Huda Diab Abdulgalil: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Formal analysis.
Otman A. Basir: Writing – review & editing, Visualization, Validation, Supervision, Funding acquisition.

Declaration of generative AI in scientific writing

During the preparation of this work, the authors used ChatGPT 4 to review the text readability. After using this tool/service, the authors reviewed and edited the content as needed, and take full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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